

Neural Network-Based Predictions of Supersymmetric Contributions to the Anomalous Magnetic Moment of the Muon (Abstract)

Jordán Daniel Santillán Morales^a

^a*Research Center in Physical and Mathematical Sciences, Facultad de Ciencias Físico Matemáticas,
Universidad Autónoma de Nuevo León, San Nicolás, Nuevo León, México*

^b*daniel.santillanmrls@uanl.edu.mx / santillanjordan9@gmail.com*

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In this work, we analyze the well-known anomaly in the muon’s magnetic moment using a flavor-extended Minimal Supersymmetric Standard Model (MSSM) with a hierarchical family structure in the trilinear scalar soft-supersymmetric terms of the Lagrangian, defined at the SUSY breaking scale. This framework introduces five free SUSY parameters within phenomenologically motivated ranges. We focus on parameter configurations for which the supersymmetric contribution to the anomalous magnetic moment is consistent with a discrepancy between theory and experiment within the range $[3.15\sigma, 4.15\sigma]$.

Using Neural Networks (NN), we develop a regression model to predict the supersymmetric contribution a_μ^{SUSY} from the five SUSY parameters. The training dataset is generated by sampling the parameter space using truncated Gaussian distributions, ensuring that the sampled values remain within the physically motivated parameter ranges. Parameter configurations satisfying the selected deviation range are then used as inputs to the neural network, while the corresponding a_μ^{SUSY} values, explicitly computed from the SUSY model, are used as target outputs for training.

The trained neural network provides satisfactory predictions of a_μ^{SUSY} , achieving a coefficient of determination $R^2 \approx 0.91\text{--}0.93$ with $MSE \sim 10^{-21}$ and $MAE \sim 10^{-11}$. A second network trained via transfer learning on the final layers yielded only a marginal improvement in predictive performance, but reduced the evaluation time from minutes, required by the analytical SUSY code, to approximately three seconds per prediction. Once trained, the model can be stored and reused, allowing rapid predictions without retraining. This demonstrates the potential of neural networks as a complementary computational tool for phenomenological studies and repeated evaluations of SUSY contributions. Future work includes larger datasets, Bayesian Neural Networks, and deeper architectures to further improve generalization.